ITCS348 Introduction to Natural Language Processing

Project 1: Sequence Classification

No more than 3 students per group

Due Date (Report + VDO presentation): Sat. 28 February 2026
Evaluation Date: 3 – 16 March 2026

Asst. Prof. Charnyote Pluempitiwiriyawej, Ph.D.
charnyote.plu@mahidol.ac.th

Goal & Objectives

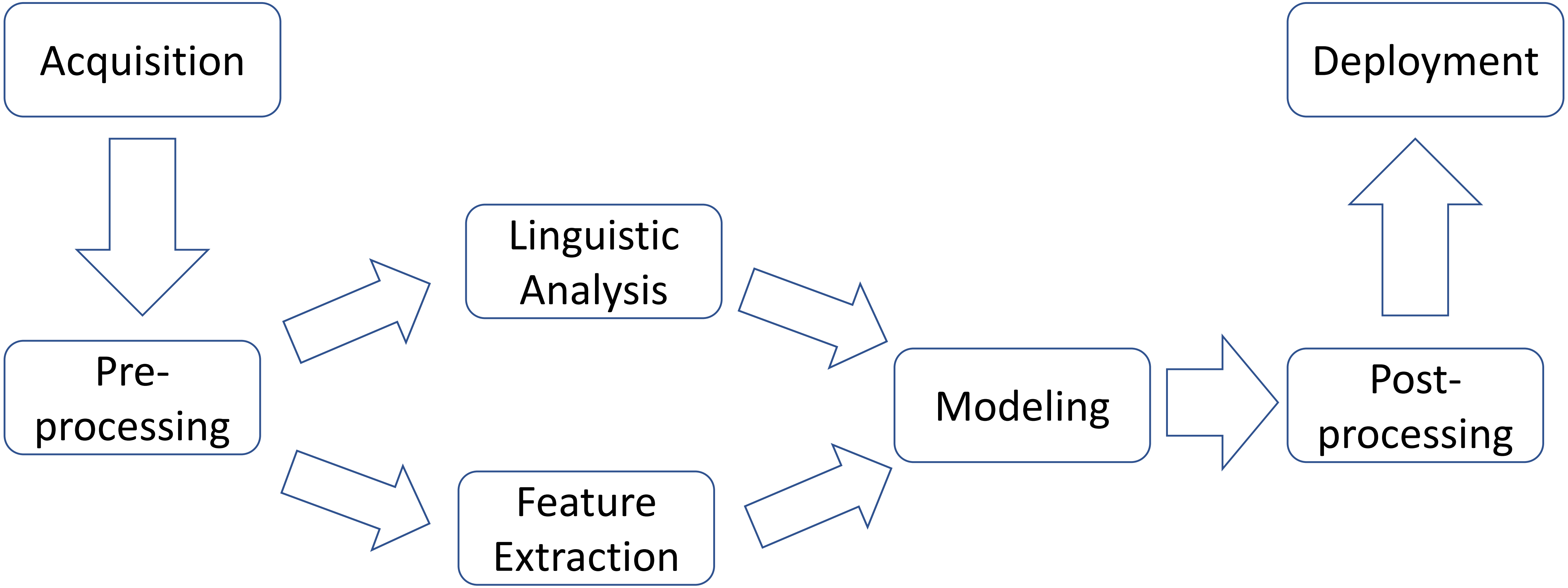
- Apply knowledge about text pre-processing and text classification techniques.
- Design, implement, and evaluate multiple text classification models on a selected dataset, comparing performance across different approaches

Learning Outcomes

- By the end of the project, students will be able to:
  - Preprocess and prepare raw text data for classification.
  - Apply and compare classification approaches.
  - Analyze performance metrics and interpret model behavior.
  - Communicate findings effectively through documentation and presentations.

Recall: NLP Pipeline

NLP Pipeline



Main Steps

1. Problem and dataset acquisition
2. Preprocessing
3. Feature extraction
4. Model selection
5. Model training and evaluation

1. Problem and dataset acquisition

- Select domain or area of interest,
  - E.g., sentiment analysis, spam detection, etc.
- Set clear objectives and performance targets.
  - E.g., classify news articles into categories like politics, sports, tech.
- Specify the classification type:
  - E.g., binary, multi-class, or multi-label.
- Acquire a relevant dataset.
  - E.g., Corpora format: plain text, html, xml, etc.
- Perform exploratory data analysis (EDA)
  - E.g., label distribution, document lengths, vocabulary richness

2. Preprocessing

- Cleaning tasks
  - Lowercasing / Uppercasing
  - Removing punctuation, html tags, stopwords, etc.
  - Tokenization
  - Stemming
  - Lemmatization
  - Text normalization
- Cleaning techniques:
  - Regular expression
  - Edit distance
  - Byte Pair Encoding
  - WordPiece

3. Feature Extraction

- Techniques e.g.,
  - Set of words
  - Bag of words
  - Tf-idf
  - Pre-trained word embedding
  - Pre-trained contextual embedding

4. Model selection

- Rule-based models (baseline)
- Traditional models (baseline)
  - E.g., Naïve Bayes, Logistic Regression, SVM
- Deep learning models
  - E.g., FNNs, RNNs, LSTMs
- Transformer-based models
  - E.g., BERT, RoBERTa, WangchanBERTa

5. Model Training and Evaluation

- **Split data**: Train / Validation / Test
- **Metrics**:
  - Accuracy, Precision, Recall, F1-score
  - Confusion Matrix
- Consider **cross-validation** for smaller datasets
- **Hyperparameter Tuning**
  - Use grid search or randomized search for ML models
  - Apply learning rate adjustments, layer freezing, dropout, etc. for deep models
- **Error Analysis**
  - Examine misclassified samples
  - Identify patterns in errors (e.g., ambiguous language, sarcasm)
- **Interpretability** (Optional but Encouraged)
  - Use SHAP, LIME, or attention visualization (for transformers)
  - Highlight what drives model decisions

Suggested Timeline

Week	Task
1	Dataset selection, literature review
2–3	Data preprocessing & baseline model
4–5	Implement DL/Transformer models
6	Evaluate models & conduct comparative analysis
7	Report writing & visualization
8	Final presentation/submission

Deliverables

- A report (6 – 10 pages, PDF) for peer evaluation.
- A slide (PDF) for presentation.
- VDO. Presentation
  - 7 - 10 minutes
  - Introduction of your classification task and dataset(s).
  - Your NLP pipeline: preprocessing, modeling, and classification techniques.
  - The experiment results.
  - A demonstration of your classification result (show interesting cases for qualitative analysis).
  - Conclusion.

Text-Classification Project Evaluation Rubrics

Criteria	Excellent (4)	Good (3)	Fair (2)	Needs Improvement (1)
1. Problem Understanding	Clear definition of task; includes classification type, goals, relevance	Task mostly clear; some link to NLP concepts	Vague definition; minimal relevance to NLP	Task unclear or irrelevant
2. Dataset Handling & EDA	Well-chosen dataset; thorough cleaning and insightful EDA	Adequate dataset use; basic EDA	Minimal preprocessing; limited data insight	Dataset poorly chosen or unexplored
3. Text Cleaning Approaches	Effectively addressed language-specific issues (e.g. code-switching, Thai tokenization, emoji handling).	Robust and task-specific cleaning (e.g. stopwords, punctuation, casing, noisy patterns) with clear justification.	Applied standard cleaning steps correctly with basic explanation.	Cleaning is poorly done or missing altogether.
4. Feature Selection Strategy	Clear rationale for feature types (BoW, tf-idf, embeddings, etc.) tied to problem goals.	Selected features relevant to task with some justification.	Chosen features are basic or arbitrary; rationale unclear.	Feature selection poorly planned or irrelevant.

Text-Classification Project Evaluation Rubrics

Criteria	Excellent (4)	Good (3)	Fair (2)	Needs Improvement (1)
5. Method Implementation	3+ models correctly implemented (ML + DL + Transformers); structured experiments	2 models implemented with some comparative structure	1–2 models with unclear methodology or missing comparisons	Incomplete or incorrect implementation
6. Evaluation & Metrics	Strong metric use (Accuracy, F1, etc.); confusion matrix + error analysis	Metrics applied correctly; basic error discussion	Metrics incomplete or misused; limited analysis	Evaluation missing or inappropriate
7. Comparative Analysis	Deep comparison of strengths/weaknesses across models	Comparisons made with relevant observations	Limited comparisons or shallow analysis	No real comparison between techniques
8. Ethical Considerations	Addresses bias, fairness, or risks in classification tasks	Mentions basic ethical issues	Ethical discussion superficial or incomplete	No mention of ethics or fairness

Text-Classification Project Evaluation Rubrics

Criteria	Excellent (4)	Good (3)	Fair (2)	Needs Improvement (1)
9. Report Quality	Structured, clear, well-written with figures, tables, and citations	Organized report with minor issues	Lacks clarity or structure; poorly formatted	Incomplete or disorganized report
10. Presentation	Confident, clear explanation with visual aids and strong engagement	Covers main points clearly; some visual support	Basic presentation; unclear delivery	Disorganized or missing presentation
11. Interpretation & Discussion (Optional but encouraged)	Insightful interpretation; connects results to theory and practical use	Clear discussion; some linkage to NLP concepts	Superficial observations; weak theoretical grounding	Little to no interpretation of findings

Weight of evaluation

- 90% for project evaluation based on evaluation rubrics:
  - instructor reviews: 70%
  - peer reviews: 30%
- 10% for participation in peer reviews